

2 Additional material¹

Empirical research in computer science usually consists of four major steps:

1. Articulate research questions
2. Software development
3. Carrying out experiments
4. Evaluation

First, the research goal must be clarified, which is typically expressed as one or few research questions. Here, it is important that these research questions *combined* should be subject to just one “ping”: one clear, sharp idea [4]. For example, “To search for primes, does using a shared counter produce smaller runtimes than pre-partitioning the number space”?

In our case, these research questions will inevitably lead to the specification and implementation of some software. That is, naming what the software must fulfill and which tools must be used.

Yet, developing software is *not* the primary goal, as it typically just serves answering the research question. Therefore, the software needs to be run under certain circumstances, such that a series of organized experiments emerges. When investing the performance of a program, one [of several] metric[s] could be the total runtime per scenario.

Now that we have some runtime lists (one per scenario), we are faced with answering the research question *meaningfully*. If we have 10 values for using the shared counter with four threads, and 10 values for using the pre-partitioning with four threads, then we must somehow correctly answer the question “Are their means the same value just with some noise”? That is, are there any differences beyond statistical imprecision (e.g., because one is likely faster)? Just calculating the mean and the standard error *will not* answer this question, as neither a statistical model, nor confidence were established. In short, one cannot simply say “these means look stable, and have a decent enough distance” because “stable” and “decent enough” were not quantified. Thus, we need to assume a certain distribution of values and define a confidence level (0 % to 100 %). Then, we may formulate a *hypothesis* that we can test against our empirical data to discover statistically significant findings.

Yet, it should be noted that there is criticism on the approach of significance testing due to some inherent flaws, such as forcing a binary view on one’s data or causing publication biasing [7]. So, please, take this into consideration: “Significance tests are simple, intuitive and adapted to the scientific method, which is centered on formulating and attempting to prove hypotheses. BUT, there are many pitfalls that a user must be aware of. (...), far reaching precautions have to be taken to ensure validity of a statistical test. In any case, if a hypothesis is sufficiently important, it is unwise to rely on a single study in order to reach confirmation.” [5, p. 2320]

Let us express each implementation’s runtime of finding prime numbers as a random variable² $X_{\text{partition}}$, X_{counter} . Then, $\mathbb{E}(X_i)$ is their respective expected runtime (“average”), while $\mathbb{V}(X_i)$ denotes their expected variance. Now, if we measure the runtimes n times, we will obtain two n -valued vectors³ $v_{\text{partition}}$, v_{counter} of data from which we can estimate the expected value and expected variance by

¹This document is a simplification on the topics presented here, so you get a sane starting point. So, please reach out to additional literature, in order to get a thorough understanding.

²This “is a rule that assigns a numerical value to each possible outcome of an experiment” [9, p. 66]. So, here, it is a function that assigns the runtime, given an experiment.

³These are also referred to as *Statistic* which is a function of n observed random variables. This is then usually assumed to be a random variable itself. [9, p. 664]

calculating the means and variances of these vectors [9, cmp. pp. 665f]:

$$\mathbb{E}(X_i) \xleftarrow{\text{estimates}} \text{mean}(v_i) = \frac{1}{n} \sum_{k=1}^n v_{i,k}$$

$$\mathbb{V}(X_i) \xleftarrow{\text{estimates}} \text{variance}(v_i) = \frac{1}{n-1} \sum_{k=1}^n (v_{i,k} - \text{mean}(v_i))^2$$

The variance uses $\frac{1}{n-1}$ instead of $\frac{1}{n}$ in order to be *unbiased* [9, p. 666], i.e. to be on average on the target [9, p. 664]. Also, we assume that all samples are independent from one another and are identically distributed. This is called *independent and identically distributed* (i.i.d.).

Obviously, the mean and variance likely change, the more data we collect; especially since they just estimate the expected results. So, what would be “more correct”, a mean of 30.0 s ($n = 30$) or 31.3 s ($n = 40$)? Well, what if the true expected value was 30.65 s, i.e. their center? Then, even the higher sample count could not manage to shift results closer towards it. Therefore, we need to quantify our certainty. A common approach to defining certainty is via *confidence intervals* with a specific *coverage probability* $1 - \alpha$, based on a *significance level* α . The latter metric expresses the probability that the true value is contained within the interval⁴ [2, p. 226]. So, if the estimated parameter does not lie within it, some significant difference can be assumed (since it is so unlikely to get by mere chance). Typically, these probabilities are 95 % ($\alpha = 0.05$), 98 % ($\alpha = 0.02$) or 99 % ($\alpha = 0.01$). Although confidence intervals are not equivalent to hypothesis tests, as these intervals contain even more information [8, p. 516], they still enable hypothesis testing. We just use them here to have a – hopefully – attainable object representing what is likely to happen (inside the interval), and not likely to happen (outside of it), given our observations.

In a next step [9, p. 718], we have to define a so-called *null hypothesis* (H_0), i.e. usually the assumption that there is no effect at all. For this example, H_0 would assume that both means are equal:

$$H_0 : \mathbb{E}(X_{\text{partition}}) = \mathbb{E}(X_{\text{counter}})$$

Alternatively, one could say

$$H_0 : \mathbb{E}(X_{\text{partition}} - X_{\text{counter}}) = 0$$

The actual hypothesis testing aims at *trying to reject* H_0 , *not* to verify it. Hence, we try to “prove” the opposite

$$H_1 : \mathbb{E}(X_{\text{partition}} - X_{\text{counter}}) \neq 0$$

With confidence intervals, this can be achieved by checking (if it is true, we can reject it):

$$0 \stackrel{?}{\notin} \text{CI}_{1-\alpha}(\text{mean}(X_{\text{partition}} - X_{\text{counter}})),$$

where $\text{CI}_{1-\alpha}(\text{mean}(Y))$ is the $100(1 - \alpha)\%$ -confidence interval around the estimated mean of a random variable Y . If the rejection was successful, we may claim that there is a statistically significant difference between the two means. But if we fail to do so, we just have no reason to disbelieve H_0 ; it does not mean, they are the same [6, p. 148]. Hypothesis tests have two kinds of errors [9, p. 718]:

⁴This is not a completely accurate, but common description. The perspective is actually different. Since the true value is not random, but the interval is, it is rather the percentage of possible confidence intervals also covering the true value [8]. Such a difference is similar to “I am helping you” and “You are being helped by me”.

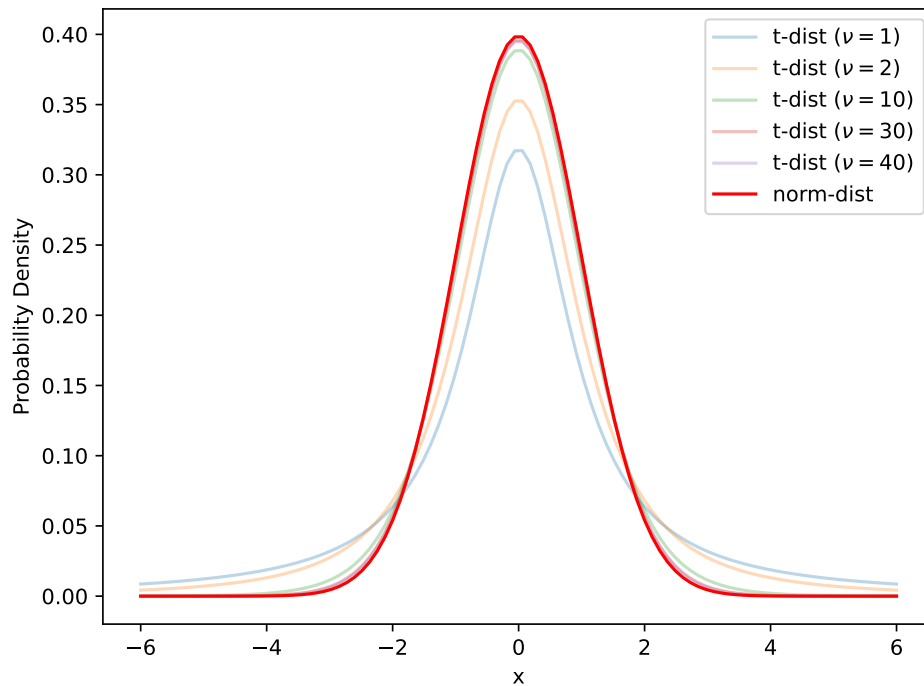


Figure 1: Standard normal distribution in relation to the t -distribution

- Type I (H_0 is actually true, but was rejected)
- Type II (H_0 is actually false, but was accepted)

Minimizing the probability of either of them for a fixed sample count will increase the probability for the other one. Only a higher sample size can decrease these probabilities at the same time [9, p. 719].

Since the coverage probability this testing builds upon is, of course, dependent on the data distribution, one must know it. To simplify things, we focus on two distributions for $X_{\text{partition}}, X_{\text{counter}}$:

1. Normal Distribution
2. (Student's) t -Distribution

Testing for the [in-]equality of means with the normal distribution bears the disadvantage of having to know the standard deviation beforehand [9, pp. 732f]. In reality, one rarely knows that in advance, which is why estimating the variance is so common. Unlike the normal distribution, the t -distribution uses a *degrees of freedom* parameter ν which modulates the “heavyness” of the tails (left and right asymptotes). For growing ν (see Figure 1), it resembles the standard normal distribution $\mathcal{N}(0, 1)$ more and more. Therefore, the t -distribution's degrees of freedom add play for an estimated standard deviation. More specifically [6, p. 140], a random variable X_i is t -distributed with $\nu := n - 1$ for

$$X_i := \frac{\sqrt{n} \cdot (\text{mean}(v_i) - \mathbb{E}(X_i))}{\sqrt{\text{variance}(v_i)}},$$

if each sample in v_i is independent from one another, and was drawn from the same *normal* distribution.

If we assume that the populations behind $X_{\text{partition}}, X_{\text{counter}}$ are independent and approximately normal, while the real variance is only estimated, one should use the t -distribution with $\nu = n_{\text{partition}} + n_{\text{counter}} - 2$ instead. Yet, this usually requires assuming that $X_{\text{partition}}, X_{\text{counter}}$ have the same – or a close-enough – distribution variance [9, p. 734]. Since this, once again, is not necessarily known, there are robust extensions, such as *Welch's Test* [1].

But what if we escalate our level of poor insight once more, so we cannot assume, our two means come from a normal distribution? Then, we can resort to *distribution-free/non-parametric* tests, such as the *rank-sum test* [9, pp. 735f]. In principle, this test arranges the concatenation of the input vectors in ascending order, such that each sample gets a global index (“rank”) in this new vector. From this, one can calculate the sum of ranks for the second input vector and obtain how likely this sum is, given the null hypothesis. Intuitively, if both vectors concentrate around different values, one rank sum will be slightly larger than the other one. If they are the same, their respective ranks will appear interleaved, hence they produce similar rank sums.

Thereby, this test does not necessarily compare means with one another, only the distributions. Thus, one has to make sure, this kind of comparison can still answer the question asked. That is, there are multiple ways, a distribution can be different from another one. For instance, if one assumes, both distributions can at most differ in their relative positioning (e.g., same variance, different mean), then a rejected test could imply a difference in means [3, p. 8]. Hence, it is still not an “assumption-free” test framework.

In general, the rank-sum test is considered preferable (presumably under assumed equal variance of populations) to the *t*-distribution-based testing, due to making fewer assumptions. However, if one is sure about the *t*-distribution, this additional information should not be wasted. [9, p. 737]

It can be non-trivial to select the correct hypothesis test, especially since there are even more available than presented here.

With Python’s SciPy library, most of the heavy lifting can be accomplished with⁵

```
scipy.stats.ttest_ind(sample_vec_a, sample_vec_b, equal_var: bool,  
                      alternative: str)
```

and⁶

```
scipy.stats.mannwhitneyu(sample_vec_a, sample_vec_b, alternative: str)
```

Both take two *independent vectors of independent samples* and operate on the null hypothesis assuming equality between a property of the two. For `ttest_ind()`, H_0 hypothesizes the equality of their respective means. Per default, it assumes, their variances are the same (`equal_var = true`), so changing it to `false` is sensible when no such claim can be made. By contrast, `mannwhitneyu()`’s null hypothesis claims that both vectors have the same distribution.

In case of both tests, one might also specify the alternative hypothesis H_1 (`alternative: str`). Next to the standard “ H_1 is the direct opposite of H_0 ”, one could also check more specifically whether one is likely larger (greater) or smaller (less) than the other. For the *t*-distribution, this is the relation between the means, and for the Mann-Whitney U rank test it is about “stochastically larger” or “stochastically smaller”.

Among these functions’ results is the `pvalue`, which can be compared to the α here. This may be used to check whether the null hypothesis should be rejected according to our defined confidence level.

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⁵https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.ttest_ind.html

⁶<https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.mannwhitneyu.html>

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